**Supplementary Material — Proofs, Noise Bounds and Reproducibility**

Companion to: Exact Support Identification of Boolean Guards from Sector Counts and Weight-Limited Queries

Supplementary technical proofs, conditional noise benchmark, exact observed-profile calculations, and reproducibility information.

## Appendix A. Proofs for Section 4

**Proof of Theorem 1.** If $f$ does not depend on the controls in a set $R$ with $|R| = r$, then $f(y) = h(y_{[m]\setminus R})$ for a guard $h$ on $k = m - r$ controls, and each word of $h$ of weight $j$ extends to $\binom{r}{i}$ words of $f$ of weight $j + i$. Hence $A_f(t) = (1+t)^r A_h(t)$, so $A_h = A/(1+t)^r = B_r$ is determined by $A$, and it exists as a profile only if the division is exact with $0 \le b_j \le \binom{k}{j}$. The guards $h$ with profile $B_r$ number $\prod_j \binom{\binom{k}{j}}{b_j} = W_r(A)$, and $h \mapsto f$ is a bijection onto the guards with profile $A$ that are independent of $R$. This count depends on $r$ only. Let $E_i$ be the event that control $i$ is inessential. The intersection over a set $R$ of $r$ events is the set of guards independent of $R$, of size $W_r(A)$. Inclusion–exclusion over the $m$ events gives the number of guards with no inessential control, $F_m(A) = \sum_r (-1)^r \binom{m}{r} W_r(A)$. For the exact support size $k$, choose the support $S$ in $\binom{m}{k}$ ways. The guards with support exactly $S$ correspond to guards $h$ on the $k$ controls of $S$ with all controls essential and profile $A/(1+t)^{m-k}$, and there are $F_k(A/(1+t)^{m-k})$ of them. Every guard has a unique support, so summing over $k$ gives all $\prod_j \binom{\binom{m}{j}}{A_j}$ guards. ▫

**Proof of Proposition 1.** The accepted words of $\mathrm{maj}(y_1, y_2, y_3)$ with $y_4$ free are three of weight two (110,101,011 followed by 0), four of weight three and one of weight four. For $g$ the accepted words are 1100, 1010, 1001 (weight two), 1110, 1101, 1011, 0111 (weight three) and 1111. Both profiles are $(0,0,3,4,1)$. The control $y_4$ is inessential for $f$ and essential for $g$, since $g(1000) = 0$ and $g(1001) = 1$, and the other three controls are essential for both. The profile has $\binom{6}{3}\binom{4}{4}\binom{1}{1} = 20$ guards. Dividing by $(1+t)$ gives $(0,0,3,1)$, which is the profile of the majority function on three controls only, so $F_3 = 1$ and $E_3 = 4$, and $E_4 = 20 - 4 = 16$. The

words of smallest weight on which $f$ and $g$ differ have weight two, for instance 0110 where $f = 1$ and $g = 0$. ▫

**Proof of Proposition 2.** For fixed $v$ the accepted $u$ form the set $\pi^p(F)$, which has the weight enumerator of $F$, namely $1 + t + t^2 + t^3 = (1 + t)(1 + t^2)$, because $\pi$ preserves weight. Summing over the $2^{m-3}$ words $v$ with weight enumerator $(1 + t)^{m-3}$ gives $A_{f_m} = (1 + t)^{m-2}(1 + t^2)$. The guard $g_m$ accepts $y_1 = y_2$, with enumerator $(1 + t^2)(1 + t)^{m-2}$. The guard $g_m$ depends on two controls. For $f_m$, controls $y_1, y_2, y_3$ are essential by the pairs 000/100, 000/010 and 010/011 at $p = 0$, and a later control $y_i$ is essential by toggling it with $u = 001$ and the other later bits at 0, since $f_m = 1$ for $p = 0$ and $f_m = 1_F(100) = 0$ for $p = 1$. The two guards agree at weight zero, and at weight one (the words $e_1, e_2$ are rejected by both and $e_3, e_i$ are accepted by both), and at the word $0110 \cdots 0$ of weight two $f_m = 1$ and $g_m = 0$. ▫

**Appendix B. Proof of Theorem 2**

For clarity, the class F($m$,$j$) is defined by $j \geq 2$, $m \geq 2j+1$ and A$\ell$=0 for $\ell < j$, A$_j$=$\binom{m-1}{j}$, A$\ell$=$\binom{m}{\ell}$ for $\ell > j$. Only weight-$j$ queries can yield an unknown answer; let a=$\binom{m-1}{j}$, $b$=$\binom{m-1}{j-1}$ and $p$=$\lceil (m-1)/j \rceil$.

Let $\mathcal{F} = \mathcal{F}(m, j)$ and let $P = \binom{m}{j}$ be the number of weight-$j$ addresses (subsets of $[m]$ of size $j$). A member of $\mathcal{F}$ is determined by the set of addresses on which it equals 1, a set of size $a$. If $q < j$ every admissible query has weight below $j$, where all members of $\mathcal{F}$ agree, so nothing can be learned, and the cost is infinite since $\mathcal{F}$ contains guards with different supports (Lemma B1). Let $q \geq j$. Queries of weight different from $j$ are uninformative, so only addresses count.

**Lemma B1 (proper supports).** If $f \in \mathcal{F}$ does not depend on control $i$, then $f = f_i$ with $f_i(y) = 1[|y \setminus \{i\}| \geq j]$, and $\mathrm{supp}(f_i) = [m] \setminus \{i\}$. Every other member of $\mathcal{F}$ has support $[m]$.

*Proof.* If $f$ is independent of $i$ then $f(y) = f(y \oplus e_i)$. An address containing $i$ is obtained from a word of weight $j - 1$ without $i$, where $f = 0$, so it has value 0. An address omitting $i$ is obtained from a word of weight $j + 1$ containing $i$, where $f = 1$, so it has value 1. This determines $f$ on weight $j$ and hence everywhere, and the count of ones is $\binom{m-1}{j} = a$ as required. The guard $f_i$ depends on every control $l \neq i$, as the words $y \not\ni i, l$ of weight $j - 1$ and $y \cup \{l\}$ show. ▫

**Lemma B2 (certificates).** Let a transcript be consistent with $f_i$. It leaves $f_i$ as the only consistent member of $\mathcal{F}$ if and only if it contains all $b$ addresses that contain $i$ (the zeros of $f_i$) or all $a$ addresses that avoid $i$ (the ones of $f_i$).

*Proof.* If an unqueried zero $z$ and an unqueried one $o$ both exist, then switching $z$ to 1 and $o$ to 0 gives a member of $\mathcal{F}$ with the same counts and the same transcript, different from $f_i$, and it has a different support (it is either another $f_i$, or has full support). If all zeros are queried, the remaining addresses must be the ones, and the guard is fixed; symmetrically for all ones. ▫

Support-identification interpretation. Lemma B1 shows that every member of the profile class with a proper support is exactly one star guard, uniquely indexed by its missing coordinate. The alternative obtained by exchanging an unqueried zero and one differs from that star guard and therefore cannot have the same proper support: if its support is proper, it must be a distinct star guard with a different missing coordinate; otherwise its support is full. Consequently, until the certificate condition in Lemma B2 is met, the transcript remains compatible with at least two distinct supports. When the condition is met, the complete function is fixed, and hence so is its support. This establishes the equivalence needed for support, without assuming full-function identification as the objective.

**Upper bound.** Maintain the set $C \subseteq [m]$ of centers $i$ for which $f_i$ is consistent with all answers so far; initially $C = [m]$. An answer 1 at address $x$ means $i \notin x$ for a consistent $f_i$ and an answer 0 means $i \in x$. While $|C| \geq 2$ ask an unqueried address $x$ with $|x \cap C| = \min(j, |C| - 1)$ and the other elements outside $C$; one exists since $m \geq j + 1$, and it is unqueried because every previous answer left either $x \cap C = \emptyset$ or $C \subseteq x$ for the then current $C$. An answer 1 removes $x \cap C$ from $C$, and an answer 0 replaces $C$ by $C \cap x$. When $|C| = 1$, say $C = \{i\}$, query the zero-addresses of $f_i$ not yet queried; an answer 1 empties $C$. When $C = \emptyset$ the support is $[m]$, and when all $b$ zero-addresses of $f_i$ have been answered 0 the guard is $f_i$ by Lemma B2. Positive answers number at most $p$. Indeed each positive answer with $|C| > j + 1$ removes $j$ centers, so at most $\lceil (m - j - 1)/j \rceil = p - 1$ such answers occur before $|C| \leq j + 1$, and one further positive answer leaves a single center. Every negative answer lies on an address containing the true center, so there are at most $b$ of them. If the true guard is a full-support member, the answers coincide with those of some $f_i$ until the last query, which is a positive answer at a zero-address of $f_i$ that empties $C$; this address is one of the $b$, so the total is again at most $b + p$. Query existence and non-repetition. Set $k$=min($j$,|C|−1). For |C|≥$j$+1, choose $j$ elements of C. Otherwise

choose |C|−1 from C and $j$−|C|+1 from outside C; enough outside labels exist since $m$−|C|≥$m$−$j$≥$j$+1. The query has 0<|$x$∩C|<|C| and is new: an earlier answer would force C⊆$x$ or C∩$x$=∅. A positive reply leaves |C|−$k$ centers and a negative one leaves $k$ centers. The algorithm terminates either with a uniquely identified star guard or with full support once a positive answer excludes the final center. The total remains at most $b$+$p$.

**Lower bound.** Let the adversary keep $C = [m]$ and answer an address $x$ with 1 if $C \setminus x \neq \emptyset$, replacing $C$ by $C \setminus x$, and with 0 otherwise, leaving $C$ unchanged. All answers are consistent with $f_i$ for every $i \in C$, and $C$ never becomes empty. A correct algorithm must stop with $|C| = 1$ since two consistent $f_i$ have different supports, and the final transcript must satisfy Lemma B2 for the surviving $i$. Each positive answer removes at most $j$ centers, so at least $\lceil (m-1)/j \rceil = p$ positive answers occur. The final transcript contains these positive answers, which are ones of $f_i$. If it contains all $a$ ones its length is at least $a$. Otherwise it contains all $b$ zeros plus at least $p$ ones, so its length is at least $b + p$. Hence the cost is at least $\min(a, b + p)$.

Adversary invariant and positive-answer count. For a star guard with center c, a weight-$j$ query at address $x$ returns 0 precisely when c belongs to $x$, and 1 otherwise. The adversary retains a nonempty set C of star centers compatible with all answers: whenever possible it returns 1 and restricts C to centers outside $x$; when no surviving center lies outside $x$, it returns 0 and leaves C unchanged. Thus an answer 1 eliminates at most $j$ centers, and an answer 0 eliminates none. Starting with $m$ possible centers and finishing with only one requires at least $p$=⌈($m$−1)/$j$⌉ positive responses. At that point the full-support hypotheses in the profile class have not automatically disappeared. Lemma B2 requires that the transcript include either all a=$\binom{m-1}{j}$ positive addresses of the remaining star guard or all $b$=$\binom{m-1}{j-1}$ of its negative addresses. The second alternative adds $b$ negative responses to the previously forced $p$ positive responses; the first requires at least a responses in total. Therefore the worst-case length is at least min(a,$b$+$p$)=$b$+$p$ by Lemma B3, exactly the upper-bound cost.

Lemma B3 (binomial inequality). For integers $j$≥2 and $m$≥2j+1, let a=$\binom{m-1}{j}$, $b$=$\binom{m-1}{j-1}$, and $p$=⌈($m$−1)/$j$⌉. Then a≥$b$+$p$. Proof. Pascal ratios give a−$b$=$b$($m$−2j)/$j$. The assumption $m$−2j≥1 gives a−$b$≥$b$/$j$. For $n$=$m$−1 and 1≤$j$−1≤$n$−1, the binomial coefficient $b$=$\binom{n}{j-1}$ is at least $n$=$m$−1. Therefore a−$b$≥($m$−1)/$j$. As a−$b$ is integral, a−$b$≥⌈($m$−1)/$j$⌉=$p$. This proof holds uniformly,

including $j$=2 and the boundary $m$=2j+1; numerical checks are not part of the logical justification.

## Appendix C. Conditional noise benchmark for the threshold-profile class

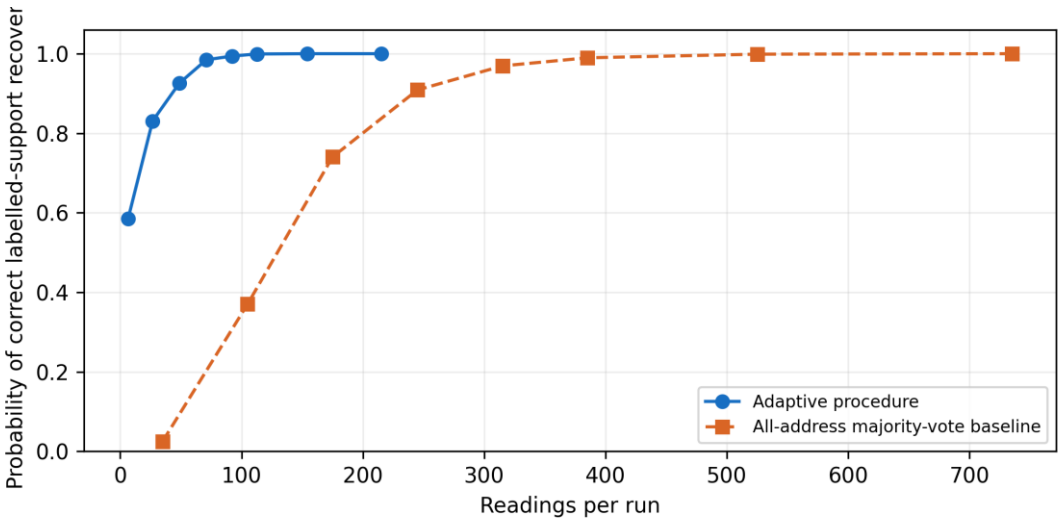
### C.1 Noisy readings for the threshold-profile class

To test the procedure outside the ideal model we simulated symmetric flips with probability $\eta$ on every reading, with each queried address read $n$ times (odd) and decided by majority. The baseline reads all 35 addresses of $(m, j) = (7,3)$ the same way and succeeds only if every address is decided correctly (an exact binomial probability), a plain-majority decoder that does not use the profile to correct errors. Each adaptive entry is the mean over 1500 runs, half of them with a proper-support guard and half with a full-support guard.

Supplement Table S1. Probability of exact support recovery under symmetric flip noise; this baseline is not the optimal profile-aware decoder.

| η | $n$ | Adaptive success | Adaptive readings | All-addresses success | All-addresses readings |
|---|---|---|---|---|---|
| 0.05 | 3 | 0.945 | 30 | 0.775 | 105 |
| 0.05 | 5 | 0.990 | 51 | 0.960 | 175 |
| 0.05 | 7 | 0.999 | 72 | 0.993 | 245 |
| 0.10 | 5 | 0.926 | 49 | 0.740 | 175 |
| 0.10 | 9 | 0.994 | 92 | 0.969 | 315 |
| 0.10 | 11 | 0.999 | 113 | 0.990 | 385 |
| 0.20 | 11 | 0.925 | 107 | 0.663 | 385 |
| 0.20 | 21 | 0.991 | 213 | 0.967 | 735 |

In the illustrative simulation, the adaptive procedure reaches success 0.990 using 51 readings at η=0.05, $n$=5, compared with 0.960 success using 175 readings for the all-address majority-vote baseline. At η=0.10, $n$=9, it reaches 0.994 success using 92 readings, compared with 0.969 success using 315 baseline readings. At η=0.20, $n$=21, it reaches 0.991 success with 213 readings while the baseline reaches 0.967 with 735 (Figure S1). The comparator is not an optimized profile-aware noisy learner; the results do not establish a minimax noisy-query advantage.




Supplement Figure S1. Probability of exact labelled-support recovery in the specified threshold-profile simulation at symmetric flip probability $\eta = 0.10$, versus mean readings per run; the comparator is the stated all-address majority-vote baseline, not an optimal profile-aware decoder.

## Appendix D. Reproducibility

Reproducibility files: the public baseline code-and-data archive contains dgsupport/core.py, tests_bruteforce.py, spectrum_all.cpp, exp_real_networks.py, exp_null_models.py and exp_certificates.py, with the source network models in data/. Exact observed-profile minimax routines and their cross-check results are in the companion observed-profile computation archive. Exact certificate results for 20 small ambiguous rules are in exact_certificates_k3_k4.csv, supported by exact_certificates_small.py and independent_milp_verify.py, with summary.json and milp_crosscheck.log. The baseline archive also contains computations from an unrelated hidden-conjunction benchmark; those results are not used in the present Supplement. All reported optimality claims must be checked against the corresponding code and scope.

## Appendix E. Exact observed-profile minimax and low-arity certificates

For each selected profile A from the published-network node table, enumerate every truth table with exactly $A_j$ enabled inputs at weight $j$. Compute its labelled support by testing all coordinate

flips. The target is the partition of this entire profile class by support, not the identity of a particular published rule.

For each query cap $q$, enumerate all eligible addressed inputs. For a nonterminal version space S, split S by the binary answer and compute the minimum, over nontrivial query splits, of one plus the larger child cost. A version space is terminal when all functions in it have the same support. An unsplittable nonterminal state has infinite cost. Two implementations are supplied: top-down memoized minimax recursion, and a separately written bottom-up dynamic program on reachable subsets ordered by cardinality.

A known-function support certificate is a set of addresses whose responses distinguish the known truth table from every profile-compatible function with a different support. The minimum is obtained by exhaustive combinations of eligible addresses. This certificate calculation is conditional on knowing the target rule; its value is not the minimax depth of a tree learning an unknown rule.

For four observed three-input profiles, the hypothesis-class sizes are all 9; the minimax costs at $q$=2 are 4,3,3,3, whereas the minimum support certificate of every compatible rule has length 2. The same values hold at $q$=3. For the observed four-input profiles (0,1,1,0,0) and (0,0,1,1,0), the class sizes are both 24. Minimax depths are respectively 6 for $q\geq2$ and 6 for $q\geq3$; $q$=2 fails for the second profile. The minimum certificates are of length 2 whenever support identification is possible. The full results and independent cross-check are stored as JSON alongside the scripts.

All values are finite exact enumerations in the specified observation model, not evidence of a universal formula for other profiles. The published network models are reused only to identify which profiles are present. The computation treats the profile, rather than the underlying published rule, as the prior information.

Coverage audit of the published corpus. Filtering results/real_networks_nodes.csv to k≥3 and nsizes>1 gives 71 ambiguous occurrences and 53 distinct pairs (k, profile). By k, the occurrence/profile counts are: k=3: 10/4; k=4: 14/9; k=5: 15/8; k=6: 10/10; k=7: 5/5; k=8: 8/8; k=9: 4/4; k=10: 3/3; k=12: 1/1; and k=13: 1/1. The six solved profiles exhaust those at k=3 and cover two at k=4. The remaining 47 distinct profiles were not subjected to full exact minimax computation. Profile counts group repeated network versions by identical layer-count vectors, without asserting statistical independence. These coverage figures are independently recomputable from the released node-level CSV.

Exact low-arity certificate audit. Among 20 distinct ambiguous published functions at k=3 or 4, exhaustive combinations and an independently implemented mixed-integer set-cover optimization both yield 13 minima of length 2, six of length 3, and one of length 7. Each matches the corresponding eight-order heuristic upper bound. The audit covers seven functions at k=3 and thirteen at k=4; it does not establish certificate optimality at larger arities. The exact target labels, profiles, certificate addresses and results are recorded in exact_certificates_k3_k4.csv. A search for the support of an unknown function is a different task with a separate minimax objective.

**Appendix F. Retrospective experimental circuit measurements**

Three source studies were used: Weinberg et al. (BLADE, 2017), Gorochowski et al. (RNA-seq, 2017), and Shis et al. (LacI/GalR, 2014). Reanalysis uses the authors' published numeric measurements where available; no numeric GFP values are inferred visually from plots.

F.1 BLADE: 113 designs yield 226 single-output Boolean functions. For each output, the measured four-state response was compared with the planned truth table after thresholds of 0.30, 0.50, and 0.70 of that output's within-circuit maximum. Exact design-table matches are 210, 161 and 121 of 226, respectively. At 0.50, the measured and designed essential supports agree in 181/226 functions; 158/226 measured supports remain invariant across the three thresholds. The threshold is exploratory and does not represent author-specified classification.

F.2 Gorochowski YFP: mean FPKM for Tube 1 through Tube 8 is 158.937, 128.522, 19581.796, 21832.624, 24455.477, 832.380, 3596.119 and 3267.051. Using the geometric midpoint (8391.572 FPKM) across the largest adjacent ratio in sorted tube means yields a three-ON/five-OFF classification; all three replicas per tube agree with the assigned binary state. The threshold is derived from these data. Since the mapping from tubes to three-bit input addresses was not established, no specific weight-layer profile is reported. Nevertheless, any three-element ON subset of a three-dimensional Boolean cube depends on all three variables, giving support {1,2,3}.

F.3 Shis et al.: three- and four-input experimental transcriptional gates are reported with GFP/OD plots and error bars. In the absence of numeric per-condition readings in the materials used, this study contributes qualitative context only.

Interpretive boundary: these are retrospective reanalyses of measured signals. They do not demonstrate a physical profile oracle, an experimental adaptive minimax protocol, or a net

measurement-cost advantage. The source spreadsheets and reanalysis files are retained in the associated experimental audit package.

## Appendix G. Full-profile adaptive versus nonadaptive support identification

**G.1 Definition. For $n\geq5$, $F_n$ is the complete class whose positive Hamming-layer counts are (0,0,1,2,1,0,…). All $n$ inputs are labelled. The learner receives this exact profile and must recover only the essential-variable support by exact membership queries.**

G.2 Theorem. Set $M_k=\binom{n}{k}$ and $r=\min(4,n-2)$. Then $Q_{NA}(F_n)=M_2+M_3+M_4-3$ and $Q_A(F_n)\leq M_2+M_4-2+\min(2r,r+2)$. Hence the additive gap is at least $M_3-1-\min(2r,r+2)$, which tends to infinity.

G.3 Nonadaptive proof, expanded. Here $L_k$ be the set of $n$-bit addresses of weight $k$, for $k$=2,3,4. The upper bound queries every address in $L_k$ except one; the single omitted response is deduced from the known number of positives in that layer. For the lower bound fix distinct a,$b$ in $L_k$. Construct an axis-aligned two-dimensional subcube $S=\{u,u+e_i,u+e_j,u+e_i+e_j\}$, where $|u|=2$ and i,$j$ are outside supp($u$), containing a and excluding $b$. If $k$=2, take $u$=a; if $k$=4, take $a=u+e_i+e_j$ and choose any two coordinates of a as i,$j$. If $k$=3, select one of the two cube directions from supp(a) and one outside supp(a); in fact there are $3(n-3)$ such cubes through a (or, equivalently, $3(n-3)$ choices of the other weight-3 vertex), and at most one of these can contain a specified different $b$ in its same layer. Since $n\geq5$, a cube excluding $b$ exists. Here $f$ is the indicator of S, and $g$ is the indicator of the set $(S\setminus\{a\})\cup\{b\}$. Both functions have one positive address in layer 2, two in layer 3, one in layer 4, and agree on every address except a,$b$. The support of $f$ has exactly $n-2$ elements, since i,$j$ are inessential. If a Boolean function on $n$ inputs with exactly four positive addresses had at least two inessential coordinates, its positive set would be invariant under both independent coordinate flips, hence must consist of exactly one entire two-dimensional axis-aligned subcube. But three corners of S uniquely determine its fourth; replacing a by $b\neq$a produces a positive set that is not a two-cube. Hence $g$ has at most one inessential coordinate and support different from $f$. Any fixed query set omitting both a,$b$ would therefore fail to identify support. So at most one address can be omitted per layer; therefore $Q_{NA}$ is at least the sum for $k$=2,3,4 of $(|L_k|-1)$.

G.4 Adaptive strategy, expanded. First query all but one address in layers 2 and 4. Because the known positive counts in each are exactly one, these answers identify the unique positive

addresses a and $b$, including when the omitted address is positive. If coordinate i is inessential, the positive set must be invariant under toggling i. Thus $a_i=0$ (otherwise toggling the unique weight-2 positive downward would create a prohibited weight-1 positive) and $b_i=1$ (otherwise toggling the unique weight-4 positive upward would create a prohibited weight-5 positive). Consequently i belongs to $I=\{i:a_i=0,b_i=1\}$, and $|I|\le r=\min(4,n-2)$. For $i\in I$, invariance holds precisely if both $a+e_i$ and $b-e_i$ are positive: the unique weight-2 and weight-4 vertices then form, together with these two weight-3 vertices, two flip-pairs. Since the entire class has exactly two positive weight-3 addresses, no further positive vertices exist to violate invariance. One valid policy checks both corresponding weight-3 addresses for each candidate, using $\le 2|I|$ queries. A second policy checks $a+e_i$ for each $i\in I$ and checks $b-e_i$ only if that first query is positive; there can be at most two such positive first answers, so the policy uses $\le |I|+2$ queries. Repeated addresses may be cached and do not increase either bound. Selecting the cheaper policy gives $\le \min(2|I|,|I|+2)\le \min(2r,r+2)$ additional queries. The total bound is $M_2+M_4-2+\min(2r,r+2)$. This verifies all candidate coordinates, including I empty, and does not require knowledge of any block architecture.

G.5 Finite case $n=5$. The full profile class contains $\binom{5}{2}\times\binom{10}{2}\times\binom{5}{4}=2{,}250$ functions, and the nonadaptive optimum is 22. Upper bound: a depth-12 decision-tree JSON was independently checked against every compatible function, certifying $Q_A(F_5)\le 12$. Lower bound: an exhaustive memoized decision-tree search returned NO for depth 11 (and 10), yielding the computed bound $Q_A(F_5)\ge 12$. The public reproducibility package contains the tree, its SHA-256 hash, the separate positive verifier, the exhaustive search code and saved per-depth results. Thus the reported computational optimum is $Q_A(F_5)=12$; the positive certificate and exhaustive negative search are distinct forms of evidence.

**G.6 Independent numerical tests and limitations. The analytical decision procedure was exhaustively checked on all 2250 functions for $n=5$ and all 42750 functions for $n=6$. The no-block full-profile theorem differs from earlier direct-sum constructions and the threshold-profile minimax theorem; the comparison with related adaptive-learning literature is given in MAIN, Section 3.**

**G.7 Exact support-deficiency distribution.** Let $D(f)=n-|\mathrm{supp}(f)|$. Within $F_n$, $D(f)$ can only equal 0, 1, or 2. Put $T_n=\binom{n}{2}\cdot C(\binom{n}{3},2)\cdot\binom{n}{4}$. Then the exact numbers of rules with D=2, D=1, D=0, respectively, are $N_2=\binom{n}{2}\binom{n-2}{2}$, $N_1=n[\binom{n-1}{2}\binom{n-1}{3}-(n-1)\binom{n-2}{2}]$, and $N_0=T_n-N_1-N_2$. Proof: if

coordinate i is inessential, the four positive inputs form two pairs invariant under flipping i. The unique weight-2 point a and the unique weight-4 point $b$ require $i \notin a$ and $i \in b$, and the weight-3 points must be exactly $a+e_i$ and $b-e_i$. Invariance under two distinct coordinates forces the positive set to be an axis-aligned 2-cube; choose its two free coordinates and its weight-2 base, yielding $N_2$. For each fixed i, choosing a from $\binom{n-1}{2}$ and $b$ from $\binom{n-1}{3}$ determines an i-invariant rule. Those also invariant in a second coordinate are counted $(n-1)\binom{n-2}{2}$ times for that fixed i, giving $N_1$. All remaining rules have full support. At $n$=5 the counts $(N_0,N_1,N_2)$=(2160,60,30); at $n$=6 they are (42240,420,90). An independent exhaustive verifier reproduces both distributions.

## Appendix H. Independently verified bounds for unresolved corpus profiles

Among the 53 distinct support-cardinality-ambiguous Hamming-layer profiles observed in the published-network corpus, six have previously determined exact adaptive minimax costs. For each of the remaining 47, we derive lower and upper bounds under unrestricted addressed membership queries (q=k). The complete 53-row inventory, with the six exact cases flagged, is provided as profile_bounds_53.csv.

### H.1. Lower bound

Let $s_{min}<s_{max}$ be the smallest and largest attainable cardinalities of the essential support. Existence of one profile-compatible function with either support size, together with coordinate permutation invariance of the profile, implies that all $\binom{k}{s_min} + \binom{k}{s_max}$ distinct labelled supports occur. Since a depth-d binary decision tree has at most $2^d$ leaves, exact support identification requires $d \geq \lceil log_2(\binom{k}{s_min} + \binom{k}{s_max}) \rceil$. This lower bound is valid for any (possibly adaptive) noiseless strategy.

### H.2. Constructive upper bound

For a fixed profile $(a_0,...,a_k)$, a layer $j$ is mixed when $0<a_j<\binom{k}{j}$. Constant layers require no queries. In each mixed layer $j$, query all but one of its $\binom{k}{j}$ addresses. The omitted bit equals $a_j$ minus the sum of the observed bits in that layer; because the profile is feasible, this recovers the entire truth table. Hence $Q_A$ is at most the sum for $j$ among mixed layers of $(\binom{k}{j}-1)$. The strategy

is nonadaptive and works for unrestricted access $q=k$. For a weight-limited interface $q<k$, this upper bound must not be used without verifying that every required layer is queryable.

### H.3. Empirical coverage and scope

Reproduction: python compute_bounds.py, using the archived real_networks_nodes.csv dataset. The script identifies 53 distinct ambiguous cardinality profiles and marks the six profiles already solved exactly. It derives bounds for all 47 remaining profiles; lower endpoints range from 3 to 10, and upper endpoints from 8 to 8089. The large upper endpoints reflect guaranteed full-table reconstruction, not typical cost or optimal strategy. The lower and upper bounds are not generally tight.

### H.4. Independent exact-spectrum audit

An independent implementation recomputed the exact support-cardinality distribution for each of the 53 profiles (including all 47 nonexact minimax cases) using forward polynomial quotient coefficients and binomial inversion over inessential coordinate subsets. It agrees exactly with the archived corpus for every support-cardinality count, profile frequency, and resulting lower/upper query bound, including all $k\geq5$ cases up to $k=13$ represented in the inventory. The standalone implementation independent_spectrum_audit.py and machine-readable independent_spectra_53.csv accompany the reproducibility package. This checks the mathematical spectrum algorithm and reported derived quantities without enumerating every truth table in large dimensions. It does not determine exact adaptive minimax costs.